Build with AI

H2S HACK2SKILL

Solution Challenge

JanHelp
Civic Technology Platform

JanHelp

AI-enabled civic complaint platform

Multilingual intake

Gemini proof verification

Team Details

JanHelp

a. **Team name:**
b. **Team leader name:** Kartik Bhalodiya
c. **Problem Statement:**

Problem Statement

How might we help citizens report civic issues in their own language, verify proof with AI, and route complaints to the right department faster for transparent urban grievance redressal?

Build with AI

Solution Challenge

H2S HACK2SKILL

Brief about your solution

JanHelp
Civic Technology Platform

JanHelp

Accessible urban grievance redressal

- Citizens report issues in English, Hindi, or Gujarati through a guided mobile app.
- Google Gemini 1.5 Flash verifies uploaded civic evidence before submission.
- The backend detects duplicates, routes the case to the nearest department, and tracks SLA progress.

Solution Summary

JanHelp combines a Flutter citizen app, a Django REST backend, and AI-assisted verification for smarter urban complaint reporting. Citizens submit issues with location and media while the platform validates evidence, detects duplicates, routes each case to the nearest department, and adds live tracking and city

Citizen-first UX
Low-friction reporting on mobile

AI + Governance
Gemini proof checks + workflow routing

Cloud-ready
Vercel deployment + managed services

Opportunities

a. How different is it from any of the other existing ideas?
b. How will it be able to solve the problem?
c. USP of the proposed solution

Answer
Unlike static complaint forms, JanHelp combines multilingual guided intake, AI proof screening, duplicate suppression, and geo-based routing in one workflow.

Answer
The platform improves report quality, reduces wrong-category submissions, prevents repeated tickets, and sends each complaint to the nearest relevant department faster.

Answer
USP: a citizen-friendly mobile journey backed by Google AI verification, departmental SLA tracking, and city-scale analytics for accountability.

List of features offered by the solution

Multilingual AI intake
Users describe issues in English, Hindi, or Gujarati.

OTP-secured onboarding
Email OTP based login and role-aware access control.

Gemini proof verification
Images are checked for category-relevant civic evidence.

Dynamic categories
Categories and subcategories load from managed backend data.

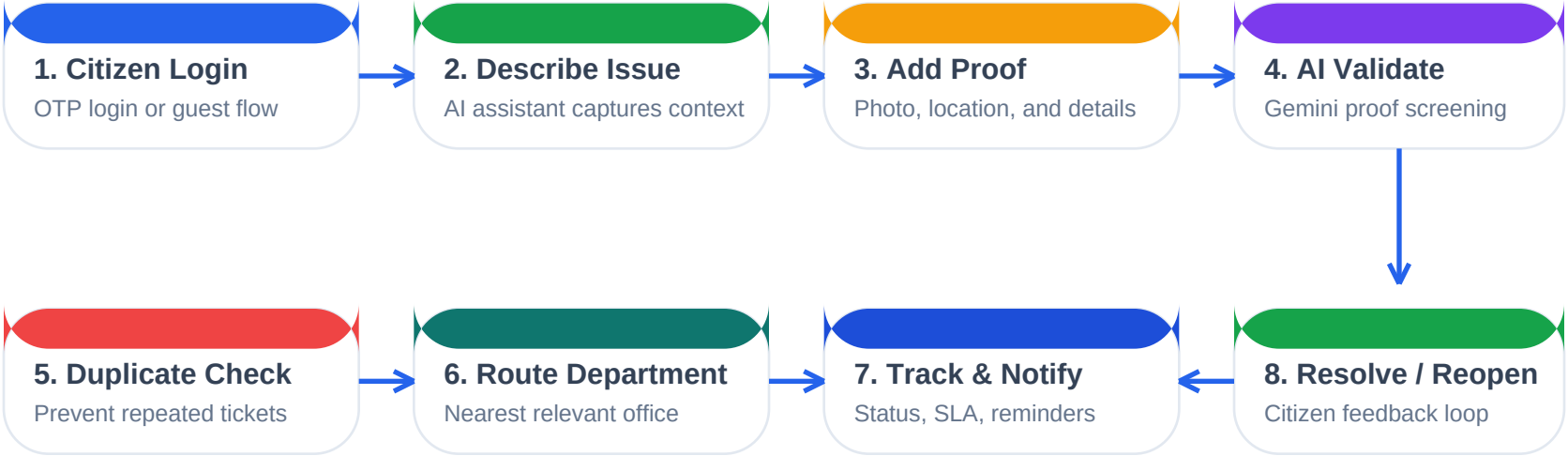
Duplicate detection
Nearby repeat complaints are suppressed using geo-radius rules.

Nearest department routing
Complaints auto-assign based on type, city, state, and distance.

Tracking, SLA, reopen
Citizens monitor status, resolution windows, and follow-up actions.

Analytics and notifications
Department workflows, heatmaps, reminders, and status alerts.

Process flow diagram or Use-case diagram



Outcome: cleaner complaints, faster routing, fewer duplicates, and better citizen trust.

Architecture diagram of the proposed solution

Citizen Layer

- Flutter mobile app
- Dashboard + tracking screens
- AI chat and complaint submission

Application Core

- Django REST API
- Authentication, OTP, JWT
- Complaint workflow engine
- Category / department services
- Tracking, SLA, reopen logic

Stakeholder Layer

- Department dashboards
- City admin analytics
- Heatmaps and complaint views
- Operational notifications

AI Services

Google Gemini 1.5 Flash proof verification, SmartCityAI extraction, duplicate heuristics, and routing decisions.

Data Layer

Supabase PostgreSQL, complaint metadata, categories, departments, and session/cache support.

Cloud Services

Vercel deployment, Cloudinary media storage, email services, and push-ready notification support.

Technologies to be used in the solution

Mobile & UX

- Flutter
- Dart
- Provider
- SharedPreferences

Backend & APIs

- Python
- Django
- DRF
- JWT

AI & Intelligence

- Google Gemini 1.5 Flash
- SmartCityAI
- NLP Extraction
- Duplicate Logic

Cloud & Data

- Vercel
- Supabase PostgreSQL
- Cloudinary
- Redis / Cache

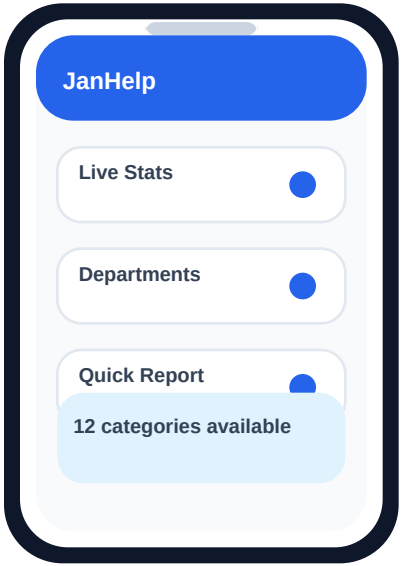
Maps & Alerts

- Geolocation
- Map-ready services
- Resend + Push Alerts

Cloud deployment + Google AI are already reflected in the current stack.

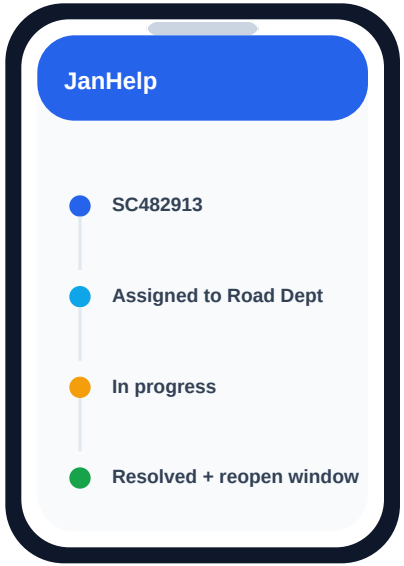
Snapshots of the MVP

Home Dashboard



AI Assistant

Tracking Flow



Other implemented screens: OTP login, category selection, proof upload, location picker, and complaint submission.

Additional Details/Future Development (if any)

Near-Term Development

- Voice-first complaint filing using speech-to-text for low-literacy users.
- Support for more Indian languages and localized assistant prompts.
- Faster department escalation when SLA breaches are predicted.
- Expanded feedback loop with post-resolution quality scoring.

City-Scale Vision

- Heatmap-based hotspot detection for ward-level planning and preventive maintenance.
- Cross-department orchestration for multi-issue complaints in the same locality.
- Citizen-to-admin insights dashboard for budget and service prioritization.
- API integrations with municipal ERP, helpline, and WhatsApp / IVR channels.

JanHelp is designed to start as an MVP and scale into a city operations platform.

Build with AI

Solution Challenge

H2S HACK2SKILL

Provide links to your:

1. **GitHub Public Repository**
   https://github.com/kartkbhalodiya/Smart-CITY

2. **Demo Video Link (3 Minutes)**
   Add your 3-minute YouTube or Drive demo link here

3. **MVP Link**
   https://janhelp.vercel.app

4. **Working Prototype Link**
   https://janhelp.vercel.app